Example 3.3

Multiply $3x^2y$ and $(2x^3y^3 - 5x^2y + 9xy)$

Think

Why $3+(4x-7y) \neq 12x-21y$?



Solution:

Now, $(3x^2y) \times (2x^3y^3 - 5x^2y + 9xy)$

$= 3x^2y(2x^3y^3) - 3x^2y(5x^2y) + 3x^2y(9xy)$

multiplying each term of the polynomial by the monomial

$= (3\times 2)(x^2 \times x^3)(y \times y^3) - (3\times 5)(x^2 \times x^2)(y \times y) + (3\times 9)(x^2 \times x)(y \times y)$

$= 6x^5y^4 - 15x^4y^2 + 27x^3y^2$

Example 3.4

Ram deposited 'x' number of ₹2000 notes, 'y' number of ₹500 notes, 'z' number of ₹100 notes in a bank and Velan deposited '$3xy$' times of amount of what Ram had deposited. How much amount did Velan deposit in the bank?

Solution:



Amount deposited by Ram

$= (x \times ₹2000 + y \times ₹500 + z \times ₹100)$

$= ₹(2000x + 500y + 100z)$

Amount deposited by Velan $= 3xy$ times $\times$ Amount deposited by Ram

$= ₹\, 3xy \times (2000x + 500y + 100z)$

$= (3\times 2000)(x \times x \times y) + (3\times 500)(x \times y \times y) + (3\times 100)(x \times y \times z)$

$= ₹(6000x^2y + 1500xy^2 + 300xyz)$



Try these

Multiply

(i) $(5x^2 + 7x - 3)$ by $-4x^2$ (ii) $(10x - 7y + 5z)$ by $6xyz$

(ii) $(ab+3bc-5ca)$ by $3a^2bc$ (iv) $(4m^2 - 3m + 7)$ by $-5m^3$

3.2.3 Multiplication of two binomials

Consider that a rectangular flower bed whose length is decreased by 5 *units* from the original length and whose breadth is increased by 3 *units* to the original breadth. What is the area of the rectangular flower bed?



Area of the rectangle $= l \times b$

Here, area of the rectangular flower bed $A = (l-5) \times (b+3)$ *sq.units*

How do we multiply this?

Now, let us learn how to multiply two binomials.

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